

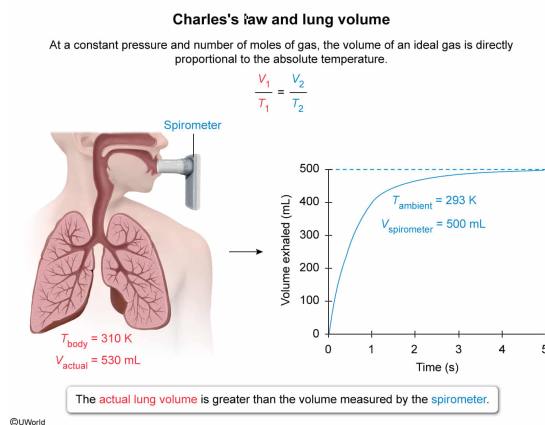
A spirometer is a machine used to assess lung volume by measuring how much air is inhaled and exhaled. If a spirometer measures a patient's lung volume to be 500 mL at an ambient temperature of 293 K, what is the actual lung volume of the patient? (Note: Assume a constant pressure of 1 atm and a body temperature of 310 K.)

- ✔ ☒ A. The actual lung volume is greater than 500 mL because of Charles's law. [59%]
☐ B. The actual lung volume is less than 500 mL because of Boyle's law. [13%]
☐ C. The actual lung volume is equal to 500 mL because of Charles's law. [6%]
☐ D. The actual lung volume is greater than 500 mL because of Boyle's law. [21%]

Correct

58% answered correctly

Explanation:



Based on the **kinetic molecular theory** of gases, the average kinetic energy of gas particles in a system depends on the temperature: As the temperature increases, the average velocity of the gas particles increases. To maintain a constant pressure as the temperature increases for a given number of gas particles, the volume of gas must also increase.

Accordingly, **Charles's law** states that a **direct relationship** exists between **temperature** T (in **Kelvin**) and **volume** V of a fixed number of particles of an ideal gas at constant pressure, according to the following equations:

$$V \propto T$$
$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

where initial conditions are indicated by the subscript "1" and final conditions are indicated by the subscript "2". These equations show that as the temperature increases while the pressure remains constant, the volume of gas also increases.

In this question, a patient's lung volume is measured to be 500 mL using a spirometer. Because the temperature of the air exhaled from the patient's lungs into the spirometer decreases from 310 K (body temperature) to 293 K (ambient temperature), the actual lung volume is greater than the measured volume. Therefore, the actual lung volume of the patient is greater than 500 mL due to Charles's law.

Although not necessary to answer the question, substituting the given temperatures and volume into the equation and solving for V_1 yields:

$$\frac{V_{1,\text{actual}}}{T_{1,\text{body}}} = \frac{V_{2,\text{spirometer}}}{T_{2,\text{ambient}}}$$

$$V_{1,\text{actual}} = \frac{V_{2,\text{spirometer}} \times T_{1,\text{body}}}{T_{2,\text{ambient}}}$$

$$V_{1,\text{actual}} = \frac{500 \text{ mL} \times 310 \text{ K}}{293 \text{ K}} = 530 \text{ mL}$$

(Choices B and D) Boyle's law describes the inverse relationship between *volume* and *pressure* for a fixed number of molecules of an ideal gas at *constant temperature*.

(Choice C) Because the volume of a gas increases with temperature, the actual lung capacity must be greater than 500 mL because the body temperature is higher than the ambient temperature.

Educational objective:

Charles's law states that a direct relationship exists between the volume and the temperature of a fixed number of molecules of an ideal gas at constant pressure (ie, as temperature increases, volume increases).

Time spent: 0 sec

Subject:
General Chemistry

QID: 404163

System:
Thermodynamics, Kinetics & Gas Laws